

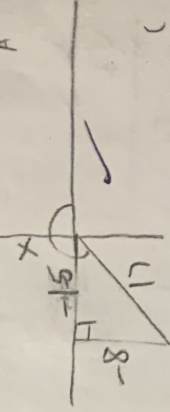
Total
21 / 21

Knowledge

1. If $\sec x = -\frac{17}{15}$, and $\pi \leq x \leq \frac{3\pi}{2}$ determine the exact values of the other 2 reciprocal trigonometric ratios. Must include a diagram for full marks.

(2) $\csc x = \frac{1}{\sec x} = -\frac{17}{15}$

$\tan x = \frac{\sin x}{\cos x} = -\frac{17}{15}$

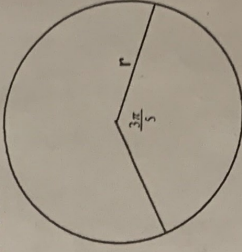


2. Complete the following conversion in the chart below. Round to two decimals where necessary.

Degree Measure	Exact Radian Measure in terms of π	Approximate Radian Measure (2 decimal places)
285°	$285^\circ \times \frac{\pi}{180} = \frac{19\pi}{12}$	4.97
$\frac{3\pi}{5} \times \frac{180}{\pi} = 108^\circ$	$\frac{3\pi}{5}$	1.88
$7.854 \times \frac{180}{\pi} = 450^\circ$	$450^\circ \times \frac{\pi}{180} = \frac{5\pi}{2}$	7.854

3. Determine the radius, to one decimal place, given the arc length to be 24 cm.

(2) $\theta = \frac{a}{r} \Rightarrow r = \frac{24}{\frac{3\pi}{5}} = \frac{40}{\pi} = 12.7 \text{ cm}$



4. A wind turbine as three blades, each measuring 4 m from centre to tip. At a particular time, the turbine is rotating six times a minute.

- a) Determine the angular velocity of the turbine in radians/s. Provide an exact answer.

(1) $\omega = \frac{6 \text{ rev}}{\text{min}} \times \frac{2\pi \text{ rad}}{\text{rev}} \times \frac{\text{min}}{60 \text{ s}} = \frac{\pi}{5} \text{ rad/s}$

- b) How far has the tip of a blade travelled after 5 min? (nearest meter)

(2) $\omega = \frac{\theta}{t}$

$\theta = \omega t$

$\theta = \frac{\pi}{5} \times 5 \text{ min} \times \frac{60 \text{ s}}{\text{min}} = 60\pi$

$a = r\theta$

$a = (4)(60\pi)$

$= 240\pi \approx 754 \text{ m}$

APPLICATION

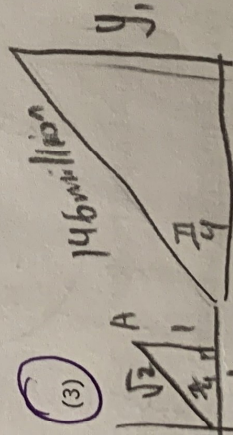
1. Use exact values to determine $\cot \frac{5\pi}{3} - \cos \frac{3\pi}{4}$. Rationalize and reduce. Show ALL work !!

	Diagram	Ratio
$\cot \frac{5\pi}{3}$		$-\frac{1}{\sqrt{3}}$
$\cos \frac{3\pi}{4}$		$-\frac{1}{\sqrt{2}}$
$\sin \frac{4\pi}{3}$		$-\frac{\sqrt{3}}{2}$

$$\begin{aligned}
 &= \frac{\left(-\frac{1}{\sqrt{3}}\right) - \left(-\frac{1}{\sqrt{2}}\right)}{-\frac{\sqrt{3}}{2}} \\
 &= \frac{-\frac{1}{\sqrt{3}} + \frac{1}{\sqrt{2}}}{-\frac{\sqrt{3}}{2}} \\
 &= \frac{\frac{\sqrt{3}-\sqrt{2}}{\sqrt{6}} \cdot \frac{2}{\sqrt{3}}}{-\frac{\sqrt{3}}{2}} \\
 &= \frac{-2(\sqrt{3}-\sqrt{2}) \times \frac{\sqrt{2}}{\sqrt{6}}}{-2\sqrt{6} + 4} \\
 &= \frac{2-\sqrt{6}}{3}
 \end{aligned}$$

2. Brett is enjoying a sunset at his cottage. As he sits outside the deck, the angle of elevation to the sun changes from $\frac{\pi}{4}$ to $\frac{\pi}{6}$. If the sun is 146 million kilometers away from Earth,

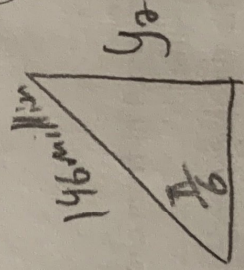
determine the exact vertical displacement of the sun over the given angle displacement. (Assume that the sun's distance from Earth remains constant). Rationalize and reduce.



$$\sin \frac{\pi}{4} = \frac{y_1}{146 \text{ million}}$$

$$y_1 = \frac{1}{\sqrt{2}} \cdot 146 \text{ million}$$

$$y_1 = 73\sqrt{2} \text{ million}$$



$$\sin \frac{\pi}{6} = \frac{y_2}{146 \text{ million}}$$

$$y_2 = \frac{1}{2} \cdot 146 \text{ million}$$

$$y_2 = 73 \text{ million}$$

$$\text{vertical distance} = |y_2 - y_1|$$

$$= |73 - 73\sqrt{2}|$$

$$= 73\sqrt{2} - 73$$

Since $y_2 < y_1$, it moved down

$\therefore$ the vertical displacement is

$73\sqrt{2} - 73$ km down or million